Product Requirements Document (PRD): Edu-Pulse

Version: 1.0

Status: Draft / Research Phase

Author: Product Architect (Agentic AI Workflow)

Target Audience: K-12 Teachers, School Administrators, Students in Low-Resource Environments

1. Executive Summary

LingoQuiz AI is a next-generation "Adaptive Assessment Platform" designed to close the gap between static classroom teaching and individual student comprehension. In regions like Nigeria, teachers face massive workloads, diverse student literacy levels, and the challenge of teaching English-centric curricula to students who process information better in their mother tongues.

LingoQuiz AI solves this by using **Agentic AI** to ingest a single piece of source material (a PDF, a voice note, or a YouTube link) and automatically generate personalized, culturally relevant, and multi-lingual quizzes. It supports high-tech classrooms (laptops) and low-tech environments (printable worksheets with QR codes and WhatsApp bots), ensuring no student is left behind due to the digital divide.

2. Problem Statement & Market Research

2.1 The Knowledge Gap

Traditional testing is "one-size-fits-all." Research shows that a student's failure in a subject like Mathematics is often actually a failure in **Linguistic Comprehension** or **Visual Processing**. If a student is a visual learner but the quiz is purely text-based, their score reflects a lack of reading speed, not a lack of mathematical ability.

2.2 Teacher Fatigue

Teachers spend an average of 10–15 hours a week on administrative tasks, including quiz creation and grading. This leads to burnout and a lack of personalized attention for struggling students.

2.3 The Linguistic Barrier

In many Nigerian schools, English is the medium of instruction, but students speak Hausa, Yoruba, Igbo, or Gbagyi at home. The cognitive load of translating the question while solving the problem leads to lower performance.

3. Target Audience Personas

3.1 The Overworked Teacher (Primary User)

- **Need:** To create high-quality assessments in minutes rather than hours.
- **Tech Literacy:** Moderate. Comfortable with smartphones but lacks time to learn complex software.
- **Goal:** To know exactly which 5 students in a class of 50 are failing and *why*.

3.2 The Mobile-First Student (Primary Beneficiary)

- **Need:** To learn in a style that matches their cognitive profile.
- **Hardware:** Often lacks a laptop; access to a family smartphone for 1 hour a day.
- **Goal:** To feel confident during testing because the questions "make sense" to them.

3.3 The School Administrator (Decision Maker)

- **Need:** Scalable tools that improve school rankings and exam results.
- **Goal:** Data-driven insights to justify teacher training or resource allocation.

4. Functional Requirements (Core Features)

4.1 Feature Set 1: The Adaptive Content Engine (ACE)

The "Brain" of the platform must perform the following:

- **Multimodal Ingestion:** Support for PDF, PNG (handwritten notes), URL (YouTube/Web), and Audio (Teacher voice notes).
- **Master Bank Generation:** AI extracts "Core Knowledge Pillars" and creates a base set of 20+ questions for every 10 minutes of content.
- **Pedagogical Re-Skinning:** * *Visual Style:* Automatically generates SVG diagrams or image-based questions.
    - *Linguistic Style:* Frames questions in story-telling formats.
    - *Logic Style:* Focuses on abstract patterns and step-by-step proofs.
- **Cultural Contextualization:** Replaces generic Western examples (e.g., "Apples") with local equivalents (e.g., "Mangoes," "Yams," "Market Stalls").

4.2 Feature Set 2: Mother Tongue Localization (MTL)

- **Bilingual Toggle:** Every quiz can be rendered as "English Only," "Mother Tongue Only," or "Side-by-Side Bilingual."
- **Supported Dialects:** Initial focus on Hausa, Yoruba, and Igbo, with an extensible framework for smaller dialects like Gbagyi.

- **Back-Translation Guardrail:** The system must translate local language responses back to English for the teacher to review, ensuring grading accuracy.

4.3 Feature Set 3: Inclusive Delivery (The "No-Laptop" Protocol)

- **QR-Worksheet Generator:** Generates a PDF where every student gets a unique version of the quiz. Each page contains a QR code linked to the Student ID and Quiz ID.
- **WhatsApp/Telegram Interface:** A light chat bot that delivers questions one-by-one, accepts text or voice answers, and provides instant feedback.
- **Vision-Based Grading:** A mobile feature for teachers to take a photo of a completed paper worksheet. The AI uses OCR (Optical Character Recognition) to extract answers and grade them instantly.

4.4 Feature Set 4: The Insight Dashboard

- **Knowledge Heatmap:** A visual grid showing concept mastery vs. learning style.
- **Intervention Groups:** AI automatically segments the class (e.g., "Group A needs a visual refresher on Algebra").
- **Parental Reports:** Simplified summary sent via SMS/WhatsApp to parents in their local language.

5. User Flow Analysis

5.1 The Teacher Flow

1. **Login:** Via Google/Microsoft Education.
2. **Upload:** Teacher drags a PDF of a textbook chapter into the dashboard.
3. **Review:** AI presents the "Master Bank." Teacher clicks "Edit" or "Approve."
4. **Style Config:** Teacher selects "Visual" and "Linguistic" adaptations.
5. **Distribute:** Teacher selects "Print Worksheets" or "Send to WhatsApp."

5.2 The Student Flow (Paper Version)

1. **Receive:** Student gets a worksheet tailored to their "Visual" profile.
2. **Complete:** Student writes answers with a pen.
3. **Submit:** Student hands the paper to the teacher.
4. **Scan:** Teacher scans with phone; student gets a score via the class leaderboard or SMS.

6. Non-Functional Requirements (NFRs)

6.1 Security & Data Privacy

- **Compliance:** Adherence to NDPR (Nigeria Data Protection Regulation) and international COPPA standards.
- **Encryption:** All student data must be encrypted at rest and in transit.
- **Anonymization:** Student names must be replaced with UUIDs before data is sent to Third-Party LLMs (OpenAI/Anthropic) to ensure privacy.

6.2 Scalability

- **Serverless Execution:** Use of Cloud Functions (via Supabase or n8n) to handle spikes in quiz submissions during exam seasons.
- **Database:** A relational database (PostgreSQL via Supabase) to handle complex relationships between Students, Concepts, and Scores.

6.3 Reliability & Maintenance (The "Architect" Model)

- **Modular Agents:** The system is built as a series of "Agents" in Antigravity. If the "Translator Agent" fails, the "Core Quiz Agent" continues to function.
- **Automated Retries:** Implementation of exponential backoff for all API calls to ensure a 99.9% uptime.
- **Low-Code Maintenance:** All logic flows are visualized in n8n/Antigravity, allowing a non-technical founder to "see" and "fix" logic without writing code.

6.4 Usability (The "Invisible Tech" Principle)

- **Loading States:** Use of "Skeleton Screens" and progress bars while AI is thinking to reduce user anxiety.
- **Zero-Config UI:** The teacher's dashboard should require zero training. Every button should have a clear tooltip.
- **Accessibility:** High-contrast modes for visually impaired students and screen-reader support.

7. Technical Stack (Non-Technical Founder Friendly)

Layer	Recommended Tool	Why?
Orchestration	**Antigravity**	Agentic IDE that allows visual building of logic.
Logic/Workflow	**n8n (Cloud)**	No-code tool to connect AI to WhatsApp, Databases, and Email.
Frontend	**React/Tailwind (v0.dev)**	AI-generated code that looks professional and modern.
Database/Auth	**Supabase**	Handles login, security, and data storage with a simple UI.
AI Models	**GPT-4o & Claude 3.5**	Top-tier models for

		reasoning and multi-lingual support.
Local LLMs	**Ollama (Llama 3)**	For processing data locally on your Zephyrus G15 to save costs.

8. Implementation Roadmap

Phase 1: MVP (Minimum Viable Product)

- Build the "Curator Agent" in Antigravity.
- Connect to a simple web-based UI generated by v0.dev.
- Enable "English-to-Hausa" translation for 5 test questions.

Phase 2: Hybrid Launch

- Implement the "QR-Worksheet" generator.
- Build the "Vision Agent" for mobile grading.
- Beta test with one school in Abuja.

Phase 3: Scaling & AI Tutoring

- Introduce the "WhatsApp Nudge Bot" for real-time hints.
- Launch the full "Knowledge Heatmap" for administrators.
- Integrate USSD support for students without internet data.

9. Success Metrics (KPIs)

- **Teacher Time Saved:** Target > 80% reduction in quiz creation time.
- **Student Engagement:** Measured by completion rates on WhatsApp vs. Paper.
- **Comprehension Lift:** Comparison of scores between "Standard Quizzes" and "Adaptive Quizzes" over a 3-month period.
- **Market Growth:** Number of schools onboarded in the first year (Target: 50).

10. Risk Assessment & Mitigation

- **AI Hallucination:** Mitigation: Always require Teacher Approval before a quiz is "Live."
- **Data Costs:** Mitigation: Use a PWA (Progressive Web App) that works offline to save student data.
- **Initial Trust:** Mitigation: Offer a "Free Tier" for individual teachers to build a bottom-up user base before selling to schools.

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